

ANALYSTLAB AFRICA

Experience Lab Internship Programme

Week 6 Project Summary**Machine Learning Engineering Track**

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Track	Machine Learning Engineering
Project	HealthConnect Clinic Experience Lab

1. Planned

Review the Week 5 pipeline for integration gaps, integrate a Data Science model or model interface, validate data flow and pipeline input/output, run error analysis on the Week 5 baseline, and complete a real (not just identified) cross-track integration activity.

2. Completed

- Error analysis on the Week 5 LogisticRegression baseline, comparing feature values across true/false positive/negative predictions.
- Trained and evaluated two comparison models (RandomForestClassifier, GradientBoostingClassifier) on the identical patient-grouped split.
- Built and validated a swappable NoShowModelInterface — a documented contract any model can plug into without touching ingestion or feature code.
- Wrote 3 passing validation tests proving the interface correctly handles bad input (empty data, missing columns) and good input identically across two different models.
- Reached out to and exchanged information with the Data Science track (Alia Al-Qadri) — a real, evidenced integration attempt.

3. Improved from Week 5

Week 5 delivered a working pipeline with one hard-coded model. Week 6 adds error analysis (why the model is wrong, not just how often), two alternative models tested on evidence rather than assumption, and — the key structural improvement — a model-agnostic interface with real input/output validation, so the pipeline is no longer tied to one specific classifier.

4. Integrated

Two different classifiers (LogisticRegression, RandomForestClassifier) were wrapped and run through the identical NoShowModelInterface calls, proving preprocessing and model components interact correctly regardless of which model is behind the interface. This interface is what a Data Science-track model will plug into once shared.

5. Track(s) Collaborated With

Data Science (Alia Al-Qadri).

6. What Was Exchanged

Alia shared her Week 5 baseline approach (LogisticRegression, patient-level split, categorical encoding + scaling, 63% accuracy, 65% no-show recall, ROC-AUC 0.68) and asked what preprocessing/model outputs the ML Engineering track would need. A prioritised reply was sent: a fitted model artifact if shareable, or her exact feature/encoding/scaling details, plus confirmation of her target definition and any known error patterns.

7. What Changed as a Result

The pipeline's model interface was built specifically to accept a model matching Alia's stated approach — the interface's required input columns match what her described preprocessing would need. As of this submission her reply is still pending, so the interface has been validated with two internally-built models instead; it is ready to receive her model as soon as it's shared, without rework.

8. Key Findings / Development Outcomes

- False Negatives and False Positives both resemble their correctly-classified counterparts on every feature — remaining model errors look like genuine behavioural noise, not a gap the current features fail to capture.
- Neither tree-based model (RF 0.672, GB 0.674) outperformed the LogisticRegression baseline (0.683) — a genuine, evidence-based result confirming the error-analysis hypothesis, not a failed experiment.
- Alia's independently-built Data Science baseline (ROC-AUC 0.68) converges closely with this pipeline's own number (0.683) — an encouraging cross-track validation signal.

9. Major Challenges

- The Data Science track's actual fitted model/artifact was not available this week despite direct outreach — worked around by building and validating the interface against internally-built comparison models instead, so integration work was not blocked.
- A markdown/code-cell mix-up in Colab (pasting formatted text into a code cell) needed correcting mid-session — a small but real reminder to check cell type before pasting.

10. Important Decisions

- Selected LogisticRegression as the Week 6 candidate model on evidence (three-way comparison), not because it was the Week 5 default.
- Chose not to modify the feature set this week, since the error analysis did not point to a specific feature gap — avoided "fixing" something that wasn't shown to be broken.
- Documented the cross-track integration honestly as pending rather than fabricating a completed handoff.

11. Remaining Issues

- Alia's actual model/artifact has not yet been received — the single largest open item for Week 7.
- The Sunday-appointment / Knowledge Base inconsistency from Week 5 remains unresolved.
- No fairness evaluation has been run on the model's use of gender/age features.

12. Contribution to the Overall HealthConnect Project

This week's work turns the ML Engineering pipeline from a single-model prototype into infrastructure other tracks can actually plug into — the validated model interface is the concrete mechanism by which the Data Science track's modelling work becomes usable inside a production-shaped pipeline, rather than staying a standalone notebook.

13. Proposed Focus for Week 7

Follow up with Alia for her model or specification; if still unavailable, reproduce her described approach from her Slack message and wrap it in the interface as a stand-in. Extend testing to the training and inference modules directly, and begin end-to-end validation of the full integrated pipeline ahead of Week 8's final presentation stage.